

# DINO CPU Project - Development Phase Overview

## Project Objective

Build a complete 8-bit CPU system from discrete logic ICs, progressing from simple sequencer to thermistor monitoring computer with 7-segment display and DAC output.

## Architecture Decisions

- **Memory Model:** Von Neumann (shared instruction/data memory)
- **Data Width:** 8-bit data path, 16-bit addressing
- **Target Application:** Thermistor temperature monitoring with digital display and analog output

## Phase Development Strategy

### Phase 1: DINO v0.0.1 - Basic Logic Sequencer

**Objective:** Build CPU-free sequential memory reader **Duration:** 2-3 weeks **Validation:** Programmable LED blink patterns

### Phase 2: Standalone ALU Validation

**Objective:** Test computational core independently **Duration:** 1-2 weeks

**Validation:** Complete ALU function verification

### Phase 3: ALU + Sequencer Integration

**Objective:** Combine instruction fetch with computation **Duration:** 2-3 weeks **Validation:** Execute mixed computational and display programs

### Phase 4: Presetable Program Counter

**Objective:** Add jump/branch capability **Duration:** 1-2 weeks **Validation:** Non-sequential program execution

### Phase 5: Enhanced Instruction Set

**Objective:** Complete CPU instruction set **Duration:** 3-4 weeks **Validation:** Full programming capability

### Phase 6: Peripheral Integration

**Objective:** Add 7-segment display and SPI DAC **Duration:** 2-3 weeks **Validation:** Numerical display and analog output

## Phase 7: Thermistor Monitoring System

**Objective:** Complete temperature monitoring application **Duration:** 2-3 weeks **Validation:** Real-time temperature measurement and display

### Documentation Standards

Each phase includes:

- Complete component BOM
- Detailed test procedures
- Logic analyzer validation points
- Success criteria definition
- Troubleshooting guide
- Integration notes for next phase

### Tools Required

- Logic analyzer (16+ channels)
- 5V power supply
- Breadboards and jumper wires
- Manual switches and pushbuttons
- LED indicators
- Basic test equipment (multimeter, oscilloscope)

### Project Timeline

**Total Duration:** 14-20 weeks **Milestone Reviews:** End of each phase **Documentation:** Continuous throughout development